01 INPUT

CIFAR-10

32 × 32 images · 10 classes



TRAINING BUDGET

1,000 | 5,000

INPUT FEATURES

RGB | PCA-64

10 draws × 2 starts

02 MATCHED INITIALIZATION

Shared vectors + image order

SOM = self-organizing map

SOM

Neighbors updated

Winner-only

Winner updated

IDENTICAL LLOYD REFINEMENT

Same objective · same stopping rule

16 prototypes per model

320 models across 4 initializers

03 OFFICIAL TEST

10,000 held-out images

QUANTIZATION ERROR

3 OF 4 conditions

Holm-significant vs winner-only

1.35%	0.80%	0.42%
1k RGB	1k PCA-64	5k RGB

CLASS AGREEMENT

ARI ↓ at 1k, both features

NMI: no Holm-significant effect

Geometry ≠ class alignment